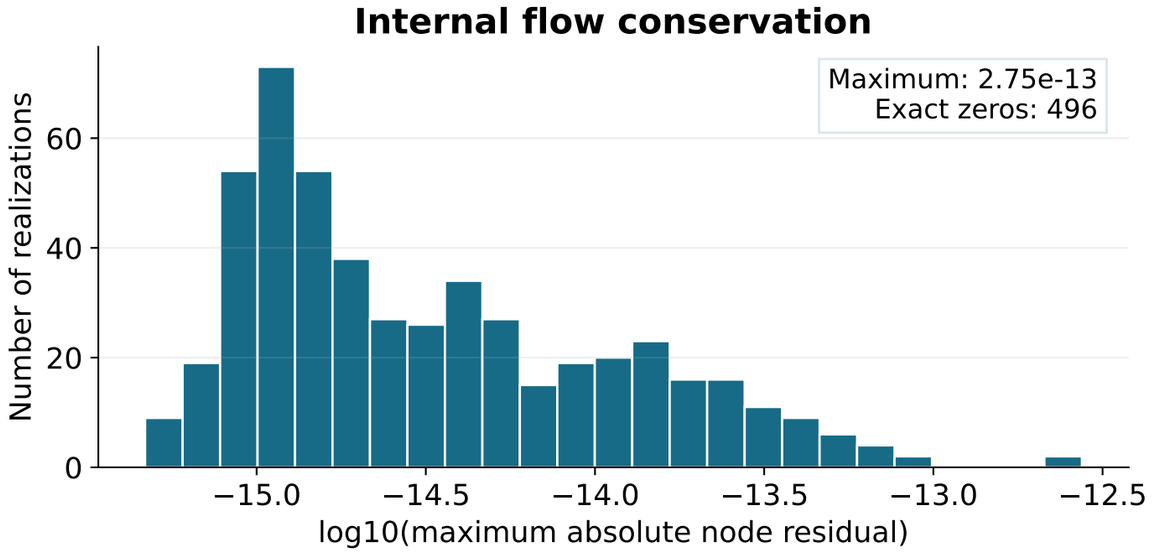
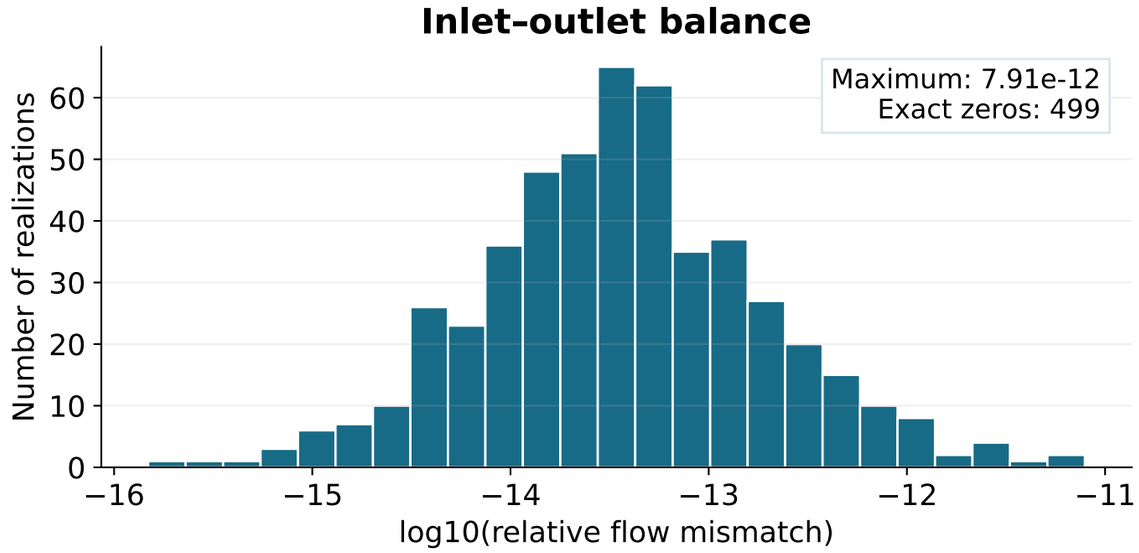
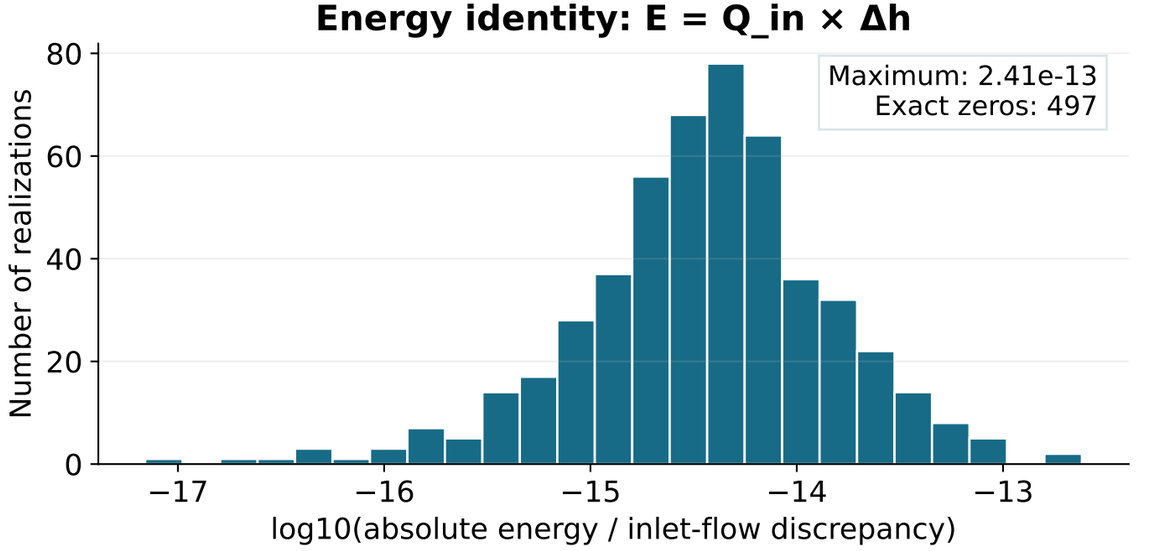
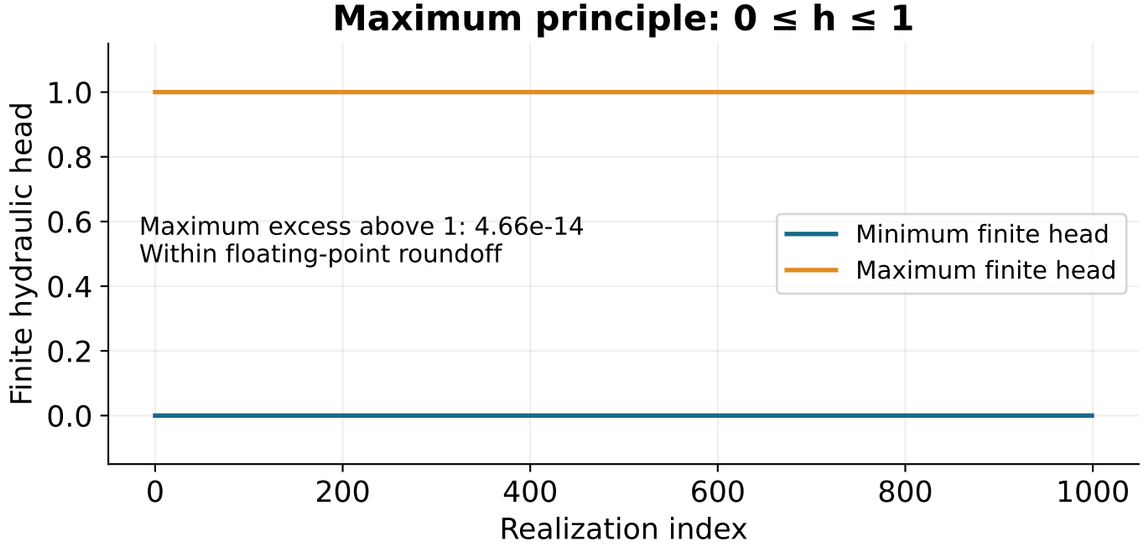
Numerical verification: conservation, energy and head bounds

1,000 independent random networks | n=25 | float64 | unit head difference





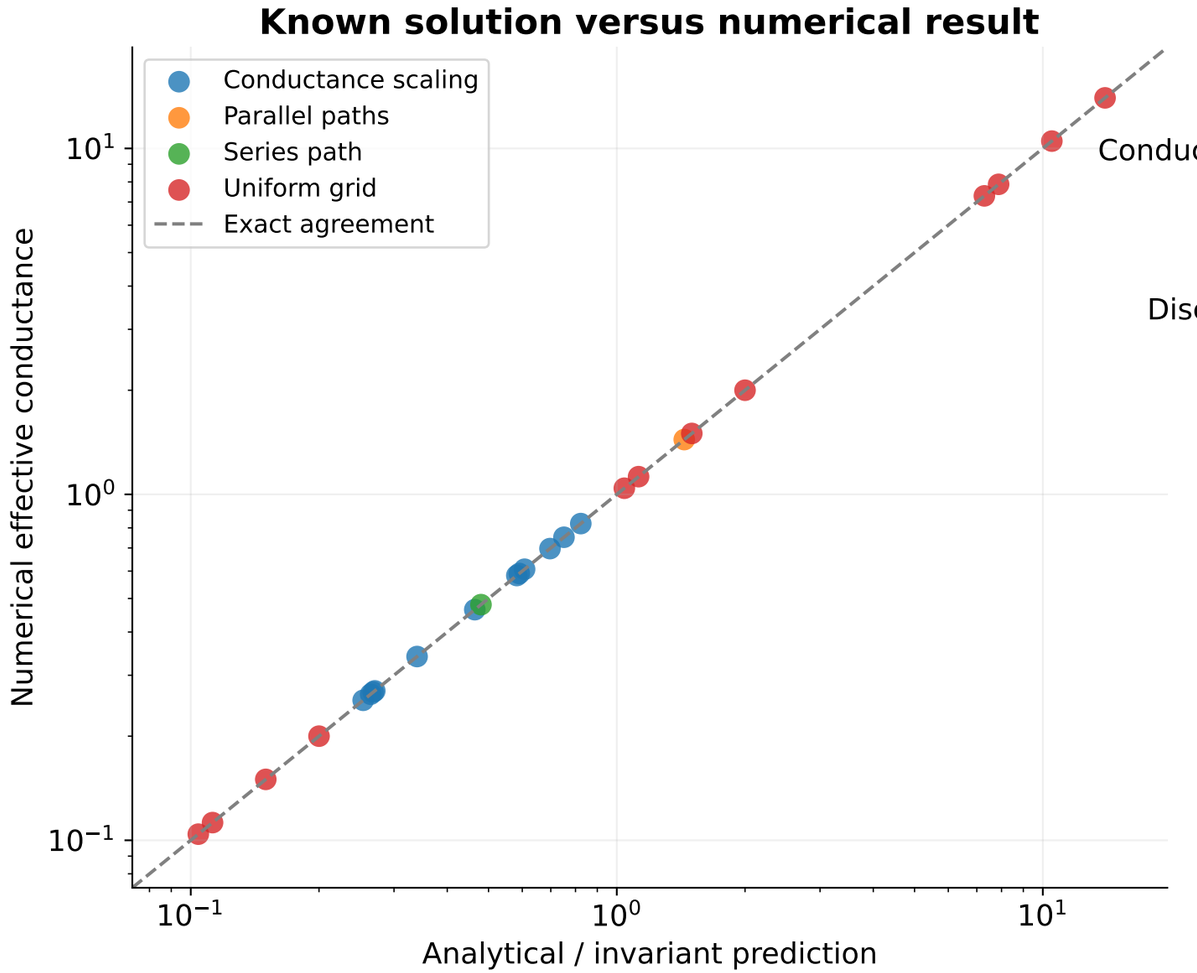


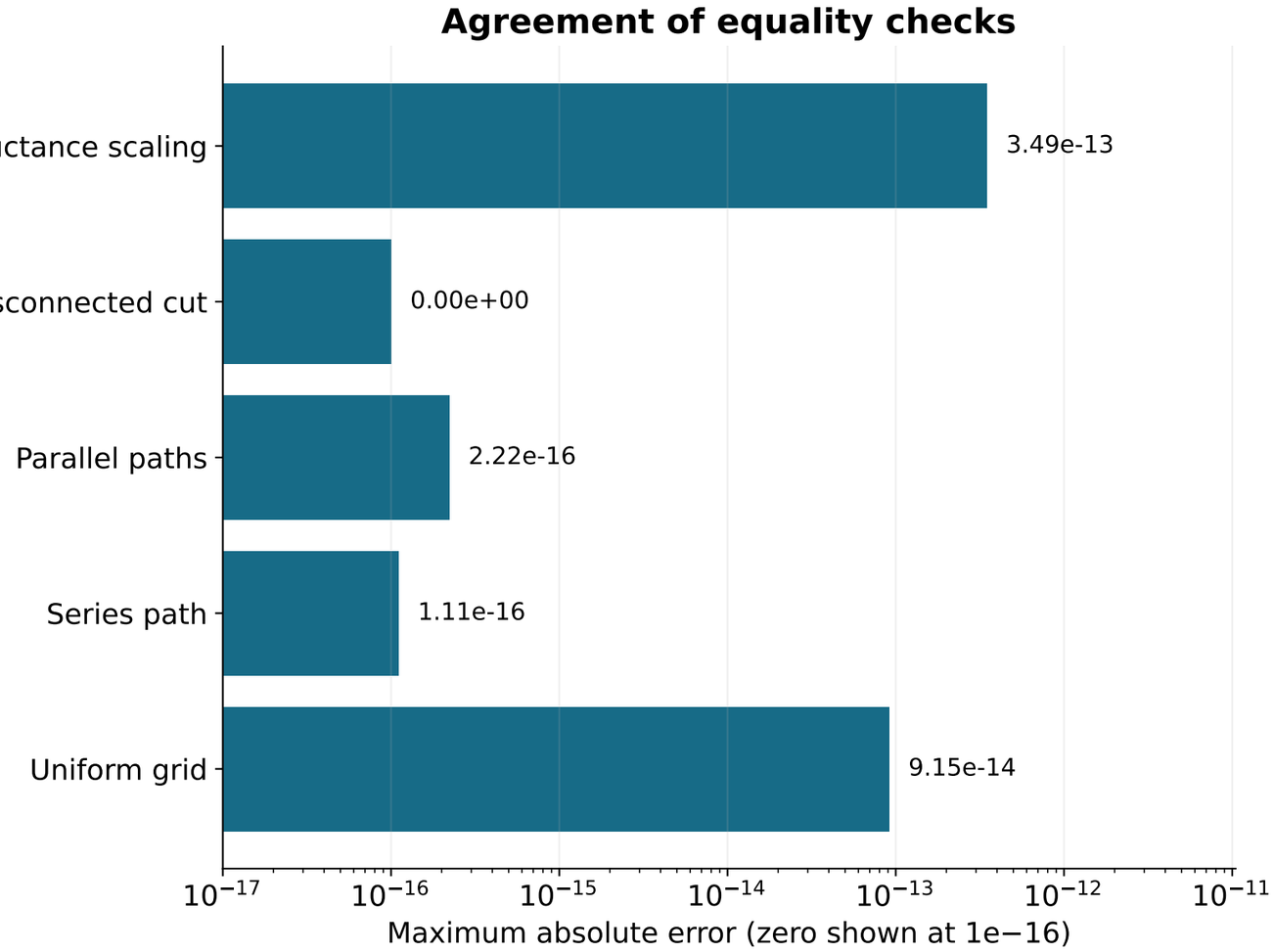


Source: results/solver_diagnostics.csv. Histograms exclude exact zeros (counts shown). Floating heads are undefined and excluded from head bounds.

Independent analytical benchmarks and physical invariants

39 recorded checks | all passed | complements the original 24 automated tests





- Uniform grid: G = g n/(n−1); linear head verified for 4 sizes × 3 conductances.
- Series: G = 1/Σ(1/g); parallel: branch conductances add; disconnected cut: G = 0.
- Scaling: G(3g) = 3G(g). Rayleigh monotonicity: increasing 30 edges never reduced G (12 seeds).

Source: reports/verification/analytical_results.csv; test definitions: tests/test_model.py. These are implementation checks, not geological field validation.